3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Detection of Microorganisms with MS: Field-Portable Instrumentation and Innovative Methodology

Authors & Institutes:

- Franco Basile, Angelo Madonna, Kent J. Voorhees (Colorado School of Mines — *CSM)*
- Stephen Lammert — *Oak Ridge National Laboratory*
- Brian Musselman, Vladimir Doroshenko (Science & Engineering Services, Inc. — *SESI)*

ABSTRACT

Abstract: The detection of biological warfare agents on the field is a challenging problem. Our research efforts in the last 17 years have contributed to the development of the Chemical Biological Mass Spectrometer (CBMS) for the US Army. The CBMS detector is a field-portable MS system based on a quadrupole ion-trap mass analyzer and a sample pre-processing unit. The entire unit is about 5.8 cubic feet, weights 170 lb, operates at 24 Vdc at less than 500 W. The sample pre-processing unit consists of a high efficiency impactor (optimized for 2-10 mm particles) and a thermal reaction chamber. Collected microorganisms from bioaerosols are enriched in this chamber and thermally hydrolyzed and methylated to produce specific biomarker molecules, a process that takes place in under 10 seconds. Subsequent mass analysis of these molecules produces biomarker fingerprints that are characteristic to that microorganism. A more detailed description of the sample pre-processing step as well as the ion trap MS hardware will be presented for the latest version of the CBMS block II system. In addition, new directions and approaches in biodetection with MALDI-MS will be presented. Briefly, affinity separation techniques are coupled with MALDI-MS analysis for the detection of bacteria in complex environmental/biological mixtures. Examples will be shown where these new methodologies are coupled with an atmospheric-pressure MALDI-MS instrument in order to simplify future hardware designs for field-portable systems.